Ceylon cinnamon does not affect postprandial plasma glucose or insulin in subjects with impaired glucose tolerance.

Previous studies on healthy subjects have shown that the intake of 6 g Cinnamomum cassia reduces postprandial glucose and that the intake of 3 g C. cassia reduces insulin response, without affecting postprandial glucose concentrations. Coumarin, which may damage the liver, is present in C. cassia, but not in Cinnamomum zeylanicum. The aim of the present study was to study the effect of C. zeylanicum on postprandial concentrations of plasma glucose, insulin, glycaemic index (GI) and insulinaemic index (GII) in subjects with impaired glucose tolerance (IGT). A total of ten subjects with IGT were assessed in a crossover trial. A standard 75 g oral glucose tolerance test (OGTT) was administered together with placebo or C. zeylanicum capsules. Finger-prick capillary blood samples were taken for glucose measurements and venous blood for insulin measurements, before and at 15, 30, 45, 60, 90, 120, 150 and 180 min after the start of the OGTT. The ingestion of 6 g C. zeylanicum had no significant effect on glucose level, insulin response, GI or GII. Ingestion of C. zeylanicum does not affect postprandial plasma glucose or insulin levels in human subjects. The Federal Institute for Risk Assessment in Europe has suggested the replacement of C. cassia by C. zeylanicum or the use of aqueous extracts of C. cassia to lower coumarin exposure. However, the positive effects seen with C. cassia in subjects with poor glycaemic control would then be lost.